

SITEX

Spatial Intelligence Platform for Café Site Selection

PROJECT CONTEXT DOCUMENT

Final Semester Execution Reference

8th Semester · Deadline: June 11, 2026

Team Members

Sujal · Roj · Subekshya · Anish

HOW TO USE THIS DOCUMENT

Paste this entire document (or relevant sections) as context when asking Claude, ChatGPT, or any AI assistant for help on any phase of this project. It contains everything needed to generate plans, code, decisions, and documentation.

1. Project Identity

Project Name: SiteX

Type: Web Application (React + FastAPI)

Domain: Spatial Intelligence / Business Analytics / GeoAI

Stage: Final Semester (8th) — continuing from 7th semester baseline

Final Deadline: June 11, 2026

Mid-Defense: ~May 11, 2026

Feature Freeze: May 30, 2026

1.1 Elevator Pitch

SiteX is a spatial intelligence platform that helps entrepreneurs evaluate whether a candidate location is commercially viable for opening a café. It analyzes the surrounding geographic context — competitor density, proximity to key Points of Interest (POIs) like schools, banks, hospitals, and temples — and generates a data-driven suitability score, a risk classification, and a natural-language explanation of why that site scored the way it did.

1.2 Core Value Proposition

- Reduces site-selection risk for café entrepreneurs using real geospatial data
- Moves beyond gut-feel decisions to graph-aware, ML-driven scoring
- Provides explainable AI output — not just a number, but a business narrative
- Identifies whether nearby competitors create synergy (destination hub) or cannibalization (market saturation)
- Adapts recommendations to the user's specific business model (premium vs. budget café)

2. Technology Stack

2.1 Frontend

Technology	Role / Notes
React + TypeScript	Main UI framework — component-based, typed
Vite	Build tool and dev server
React-Leaflet	Interactive map — pin dropping, GeoJSON layer rendering
Leaflet Heatmap Plugin	Density heatmap overlays (planned for sem 8)
Axios / Fetch	API calls to FastAPI backend

2.2 Backend

Technology	Role / Notes
Python 3.x	Primary backend language
FastAPI	REST API framework — async, auto-docs via Swagger
Pandas / GeoPandas	Data processing and geospatial operations
XGBoost	Baseline ML model (tabular, tree-based) — currently deployed
PyTorch + PyG	For HeteroGAT graph neural network (sem 8 target)
Google Gemini API	LLM for natural-language explanations (free tier)
Shapely / Fiona	Geospatial geometry processing

2.3 Data & Storage

Technology	Role / Notes
Database	NOT YET DECIDED — evaluating PostgreSQL+PostGIS vs SQLite
GeoJSON files	Spatial data for cafés, banks, POIs, roadways
CSV (master_cafes_metrics.csv)	Core training dataset with engineered features
OpenStreetMap	Source for road network (Roadway.geojson)
.pkl model files	Serialized XGBoost model: xgb_baseline.pkl

2.4 Dev Tools & Collaboration

Tool	Usage
GitHub	Version control — source of truth for all code
Trello	Task tracking and sprint management
Discord	Weekly team sync (fixed day/time, async discussions)
Jupyter Notebooks	ML training, EDA, and feature engineering experimentation

3. What Was Built in Semester 7 (Baseline)

3.1 Data Collection & Feature Engineering Pipeline

- **Scraped, cleaned, and standardized large geospatial datasets of businesses and POIs**
 - Categories: banks, cafés, education institutions, health facilities, temples
 - Output stored in CSV_Reference/ directory
- **Engineered tabular dataset: master_cafes_metrics.csv**
 - Hub-and-spoke and radius-based distance metrics to quantify neighborhood features
 - Both Euclidean distance and density-based features per POI category

3.2 Baseline Machine Learning Model

- **XGBoost tree-based model trained to predict café suitability score (continuous)**
 - Training notebook: train_xgb.ipynb
 - Exported model: models/xgb_baseline.pkl
- Input: static vector of surrounding POI distances and densities
- Output: suitability score + risk classification (Low / Medium / High)

3.3 Interactive Web Application (Frontend)

- **React + Vite frontend using react-leaflet for interactive map**
 - Users can drop a pin on any location in the city
 - Visualize POI layers via GeoJSON overlays (cafes.geojson, banks.geojson, etc.)
- Results view (Result.tsx): displays predicted score, risk level, neighborhood breakdown

3.4 FastAPI Backend

- **POST /api/v1/predict-score/ — core prediction endpoint**
 - Accepts: latitude, longitude
 - Computes tabular features on the fly using GeoPandas
 - Runs features through XGBoost model
 - Returns: suitability score, risk classification, feature breakdown
- Basic data-serving endpoints for GeoJSON and static data artifacts

3.5 Generative AI Explainability

- **POST /api/v1/explain/ — Gemini LLM integration**
 - Analyzes XGBoost model's raw numerical feature weights
 - Generates human-readable business explanation of the score
 - Highlights biggest positive and negative POI contributors
 - Uses Google Gemini API on free tier — no paid API costs

4. What Remains to Build in Semester 8

The following are the planned features for the final semester, listed in priority order. HeteroGAT is the stretch goal — if it is not production-ready by May 20, the team ships an enhanced XGBoost with graph-derived features instead.

#	Feature	Owner(s)	Priority
1	Database setup (PostgreSQL+PostGIS or SQLite)	Sujal, Roj	CRITICAL — blocks everything

2	Graph construction pipeline (POIs + Roadway.geojson → PyTorch)	Sujal, Roj	Critical
3	Road-network snapping + subgraph extraction	Sujal	Critical
4	HeteroGAT model training (or enhanced XGBoost fallback)	Sujal	Critical
5	Replace /predict-score/ with new model	Sujal	Critical
6	Frontend: road-network routing + isochrone overlays on Leaflet	Sujal	High
7	Frontend: density heatmaps as toggleable layers	Sujal	High
8	Café/retail characterization — enrich with price tier, chain vs. independent, ambiance	Roj	High
9	User persona input UI (business model: premium vs budget café)	Sujal	High
10	Persona-aware scoring backend + Gemini prompt adjustment	Subekshya, Sujal	High
11	Upgrade /explain/ endpoint for graph topology factors	Sujal	High
12	Competitor analytics: synergy vs. cannibalization metric	Subekshya	Medium
13	Automated data pipeline for new areas/cities	Roj	Medium
14	Dynamic dashboard polish (toggles, responsive, state mgmt)	Sujal	Medium
15	Full project report (all chapters)	Anish + All	Critical

5. Team Structure & Roles

Member	Primary Role	Responsibilities
Sujal	Frontend / Backend / AI (Lead)	React UI, FastAPI endpoints, ML model pipeline (XGBoost + HeteroGAT), Gemini integration, graph construction, road-network snapping. Overall technical lead.
Roj	Backend / Data Engineering	Data collection, preprocessing, POI enrichment (price tier, chain vs. independent), graph node feature preparation, automated data pipeline.
Subekshya	Backend / Analytics	Competitor analytics module, synergy vs. cannibalization scoring, persona-aware scoring backend, evaluation metrics, model performance analysis.
Anish	Documentation / Presentation	Full project report (primary author), API documentation, Trello board management, presentation slides, mid-defense and final defense preparation.

5.1 Known Constraints

- All 4 members have internships running concurrently — time is limited
- No budget for paid APIs or cloud services — all tools must be free/open-source
- Historical team issue: poor communication, missed internal deadlines, poor punctuality to submissions
- Mitigation: weekly Discord sync (fixed schedule), all decisions logged on Trello, no silent failures

6. System Architecture

6.1 High-Level Data Flow

User drops pin on Leaflet map → Frontend sends {lat, lng, persona} to FastAPI → Backend computes features (tabular or graph-based) → ML model returns suitability score → Gemini generates explanation → Frontend renders score, risk, explanation, and map overlays.

6.2 Key API Endpoints

Endpoint	Method	Description
POST /api/v1/predict-score/	POST	Accepts {lat, lng, persona?}. Returns suitability score, risk level, feature breakdown.
POST /api/v1/explain/	POST	Accepts feature weights from model. Returns Gemini-generated natural-language explanation.
GET /api/v1/pois/{category}	GET	Serves GeoJSON for a POI category (cafes, banks, etc.) for map layer rendering.
POST /api/v1/competitor-analysis/	POST	Returns synergy vs. cannibalization verdict for candidate location. (Sem 8)

6.3 ML Model Architecture

Current (Semester 7 — Deployed)

- Model: XGBoost (tree-based gradient boosting)
- Input: tabular feature vector — Euclidean distances + radius densities for each POI category
- Output: continuous suitability score (0–100) + risk class (Low / Medium / High)
- File: models/xgb_baseline.pkl

Target (Semester 8 — In Development)

- Model: Heterogeneous Graph Attention Network (HeteroGAT) via PyTorch Geometric
- Graph nodes: café locations, POIs, road intersections
- Graph edges: road connections with traversal weights (main road vs. side road vs. dead end)
- Node features: POI type, price tier, chain/independent, proximity to road centrality
- Key capability: models influence propagation — recognizes walkability via main avenue > spatial proximity across an impassable barrier
- Fallback plan: if HeteroGAT not ready by May 20, ship XGBoost with graph-derived centrality features added

6.4 Key Data Files

File	Description
master_cafes_metrics.csv	Core training dataset — engineered feature vectors for all cafés
CSV_Reference/	Raw + cleaned POI datasets by category
models/xgb_baseline.pkl	Serialized trained XGBoost model for inference
cafes.geojson	Café locations for map layer rendering
banks.geojson	Bank locations for map layer rendering
Roadway.geojson	Road network topology for graph construction
heatmap.ipynb	Reference notebook for heatmap concepts to implement in UI
train_xgb.ipynb	XGBoost training notebook — reference for model retraining

7. Project Timeline & Milestones

Phase	Dates	Key Deliverables	Hard Deadline
Phase 1: Foundation	Apr 11–25	DB decision, data migration, team setup, Trello board	Apr 25
Phase 2: Mid-Defense	Apr 26–May 11	Graph pipeline, map upgrades, heatmaps, competitor analytics, defense slides	May 11 (Mid Defense)
Phase 3: ML & Persona	May 12–30	HeteroGAT/XGBoost upgrade, persona UI, Gemini upgrade, synergy metric	May 30 (Feature Freeze)
Phase 4: Report	Jun 1–5	All report chapters written and cross-reviewed	Jun 5
Phase 5: Final Prep	Jun 6–11	Demo rehearsal, final slides, bug fixes, GitHub cleanup, submission	Jun 11 (FINAL)

CRITICAL RULE — Feature Freeze

Zero new features after May 30. After this date, only bug fixes, performance improvements, and UI polish are allowed. Report writing must START on May 12 alongside development, NOT after features are complete. Running them in parallel is non-negotiable.

8. Final Report Structure

The full project report is a required deliverable. Anish is primary author; each technical section is reviewed by its domain owner. Report writing runs in parallel with development from May 12.

Ch	Title	Primary Author	Technical Reviewer
1	Introduction — problem, motivation, objectives	Anish	Sujal
2	Literature Review — related work in GeoAI, site selection	Anish	Subekshya
3	Data Collection & Feature Engineering	Roj	Sujal
4	System Architecture (full-stack + API)	Anish + Sujal	Sujal
5	ML Model — graph construction, HeteroGAT/XGBoost, training	Sujal	Sujal
6	Analytics — competitor analysis, persona scoring, synergy	Subekshya	Sujal
7	Results & Evaluation — performance metrics, user testing	Subekshya + Anish	Sujal
8	Conclusion & Future Work	Anish	All
Ap p.	API Documentation (all endpoints)	Anish	Sujal

9. Team Operating Rules

Given the team's known issues with communication and punctuality, these rules are agreed upon by all members and apply for the entire semester.

Communication Protocol

Weekly sync: Discord, fixed day and time, 30 minutes maximum, no skipping.

Task updates go on Trello — not WhatsApp messages. WhatsApp is for social only.

If a task will miss its deadline, flag it on Discord 3 days early — no silent failures.

All technical decisions are documented on the relevant Trello card.

Sujal reviews all technical output before submission — he has the broadest context.

Scope Management Rules

HeteroGAT is a stretch goal. If not working by May 20, ship enhanced XGBoost — no debate.

Feature freeze is May 30. This date is non-negotiable regardless of internship workloads.

No new features after May 30. Only bug fixes, performance, and polish.

No paid APIs or cloud services. Gemini free tier + PyTorch + OSM data covers all needs.

10. Quick Reference for AI Assistants

When using this document as context to get AI help, append one of the following prompt openers after pasting the document. These will direct the assistant to the most relevant section.

10.1 Prompt Starters by Use Case

Use Case	Suggested Prompt Opener
Database decision	Based on section 2 and 3, recommend the best database for SiteX and explain the setup steps.
Trello sprint board	Using section 7, generate a full Trello board structure with lists, cards, and due dates for 5 phases.
Graph pipeline code	Based on section 4 and 6, write Python code to build the POI graph using PyTorch Geometric.
Gemini prompt upgrade	Based on section 6, rewrite the /explain/ prompt to handle graph topology factors.
Report chapter	Using section 8, write Chapter 3 (Data Collection) based on what Roj built.
Defense slides outline	Using sections 3, 4, and 7, create a slide-by-slide outline for the mid-defense presentation.
FastAPI endpoint code	Based on section 6.2, write the FastAPI code for /api/v1/competitor-analysis/.
React component	Based on section 4 feature #9, build the React user persona input component.
Team retrospective	Based on section 9, suggest how the team can improve weekly sync and accountability.

How to Use This Document

1. Copy this entire document (or the specific sections relevant to your task).
2. Paste it at the start of a new chat with any AI assistant.
3. Append your specific question or task after the document.
4. The AI will have full project context to give you precise, relevant answers.

Tip: For code tasks, always include Section 2 (stack) + Section 6 (architecture).

Tip: For report writing, include Section 3 (sem 7 work) + Section 4 (sem 8 work) + Section 8.

Tip: For planning tasks, include Section 5 (team) + Section 7 (timeline) + Section 9 (rules).